

Artificial Intelligence and the Brain: Is Innovation Getting Easier?

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Abstract

Does artificial intelligence (AI) make innovation easier? We develop a growth model in which AI filters accumulated knowledge into a refined knowledge load that the human brain recombines to generate new ideas. Filtering mitigates the cognitive burden of large knowledge stocks but can weaken spillovers by screening out information useful for recombination. We show that this trade-off can make research productivity nonmonotonic in the pace of AI progress. Faster AI progress raises research productivity when the effective knowledge load is low or sufficiently high but lowers it at intermediate levels. We call this intermediate region an *AI trap*: by narrowing the knowledge that reaches human recombination, stronger filtering causes lost spillovers to dominate burden relief. Along the balanced growth path, faster AI progress raises long-run growth, while its effect on the R&D labor share is ambiguous. AI can therefore accelerate growth without necessarily making innovation easier.

Keywords: AI trap, growth, knowledge spillovers, knowledge burden

JEL Codes: O40, O33, O41, O31

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1 Introduction

Artificial intelligence (AI) is becoming an increasingly important infrastructure for scientific discovery. Researchers use AI systems to search the literature, summarize evidence, write code, and navigate expanding bodies of technical knowledge. These tools have reinforced the expectation that AI will reduce the cost of innovation. Yet discovery requires more than access to information: researchers must notice, interpret, and recombine information into new hypotheses, designs, and technologies. The economic question is therefore not merely whether AI automates research tasks, but how machine processing reshapes the knowledge set on which human creativity depends.

The similarities and differences between AI and the human brain have been central to pioneering work in computer science. For example, the seminal frameworks of the Turing Test ([Turing, 1950](#)) and the von Neumann architecture ([von Neumann, 1958](#)) emphasize the imitation of brain functions by computers, laying the foundations for modern computing and AI; J. C. R. Licklider, who introduced the concept of man-computer symbiosis, argued that collaboration between computers and the brain would drive advances in formalized reasoning, scientific decision-making, and complex situation management ([Licklider, 1960](#); [Campbell-Kelly et al., 2023](#)). Recent advances in AI highlight its connection to the brain and its implications for economic development. Geoffrey Hinton, the “Godfather of AI,” stated, “[AI]’s going to be like the Industrial Revolution—but instead of our physical capabilities, it’s going to exceed our intellectual capabilities” ([BBC News, 2024](#)). However, existing economic studies have rarely examined the implications of AI’s foundational philosophy—its relationship with the brain—for innovation and growth.

In many intelligence tasks, AI has already surpassed human capabilities. For example, deep learning algorithms reduced image-labeling error rates on ImageNet—a dataset developed by Stanford researchers that comprises over 10 million images—from over 30% in 2010 to less than 5% in 2016, reaching 2.2% by 2017 and significantly outperforming the human error rate of approximately 5% ([Brynjolfsson et al., 2018](#)). In tasks involving logical and mathematical intelligence, linguistic intelligence, and natural discriminative intelligence, employing AI is substantially cheaper than hiring skilled workers ([Bao et al., 2024](#)). Furthermore, [Charness and Grieco \(2026\)](#) provide experimental evidence that AI-generated responses score higher than human-generated responses in closed tasks.

However, some scholars contend that AI alone cannot replicate human-like creativity. [Searle \(1980\)](#) maintains that intentionality emerges from the causal properties of the brain and cannot be duplicated merely

through the execution of computer programs. Consequently, Searle argues that strong AI, relying solely on programs without intrinsic causal powers, cannot achieve genuine cognition. [Felin and Holweg \(2024\)](#) argue that AI, as a data-driven predictive tool, is inherently backward-looking and imitative, lacking the theory-driven causal reasoning and forward-looking capacities intrinsic to human cognition, thus incapable of generating authentic novelty or original knowledge. [Yao et al. \(2024\)](#) highlight concerns regarding the potential marginalization of human creativity resulting from generative AI and analyze the competitive dynamics and equilibrium between human-generated and AI-generated content. [Charness and Grieco \(2026\)](#) also find that human-generated responses receive significantly higher creativity scores in open tasks; they further estimate that human imagination accounts for approximately 22 to 45 percent of the creativity score in open settings. We do not take a position on whether AI can generate autonomous novelty. Instead, we study how AI-assisted processing of existing knowledge changes the knowledge set that reaches researchers' creative judgment. In our framework, AI filters and organizes accumulated knowledge, while bounded human cognition recombines the resulting knowledge load into new production technologies.

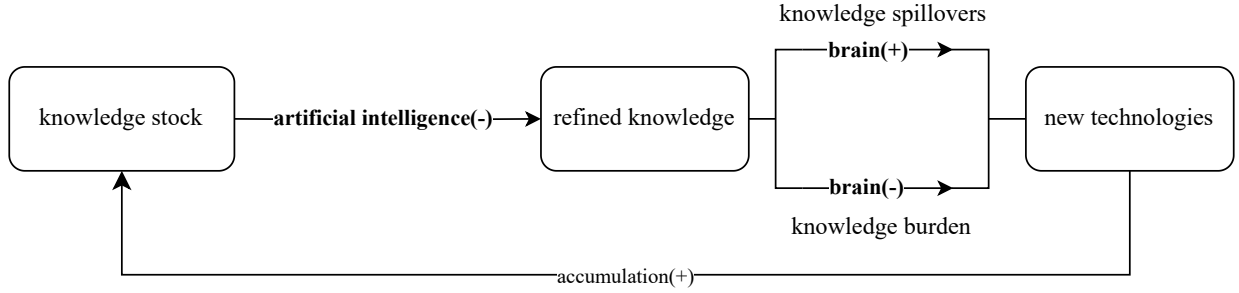


Figure 1: AI and the Brain in the Innovation Process

In this paper, we develop an endogenous growth model to capture the distinct roles of AI and the brain in innovation. In the model, “the brain” is a shorthand for bounded human cognition in the creative stage of innovation. Figure 1 illustrates the two-stage innovation process. In the first stage, AI collects, organizes, and filters existing knowledge into the refined knowledge load that reaches human cognition. Such a task cannot be accomplished by the brain alone. Each individual brain is constrained by physiological limits that hinder its ability to synthesize, analyze, and organize high-dimensional knowledge. In contrast, AI, supported by parallel processing chip architectures, is capable of handling these tasks efficiently. Even if the brain could process existing knowledge, its efficiency would be significantly lower, and relying solely on it to manage vast amounts of information is restrictive. This stage captures the idea that innovation is not exposure to all

available knowledge, but a process in which information is selected before it enters creative use.

The second stage centers on the brain’s creative processing of the refined knowledge load. This creative capacity distinguishes the brain from AI, ultimately delivering the “final blow” to generate new production technologies. At this stage, two forces arise: *knowledge spillovers* and *knowledge burden*. When the knowledge load is small, an increase expands the set of ideas available for creative recombination and facilitates innovation, as in idea-based growth models with positive knowledge spillovers (Romer, 1990; Jones, 1995). When the load becomes sufficiently large, however, it impedes creative processing and reduces innovation. This force, which we call the *knowledge burden*, has received limited attention in growth theory (Jones, 2009, 2010).

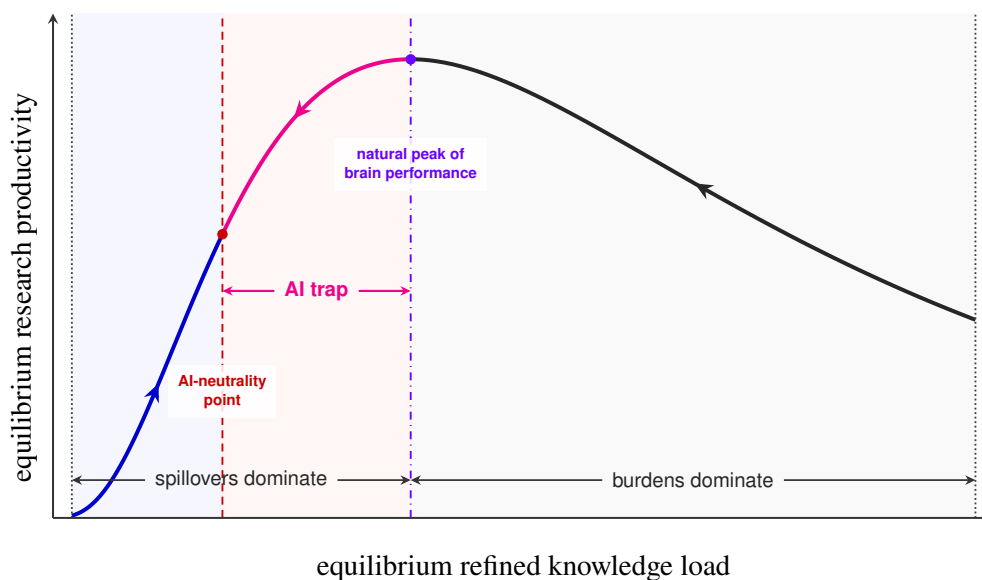


Figure 2: Effects of Faster AI Progress on Refined Knowledge Load and Research Productivity
Notes. Arrows indicate how faster AI progress shifts the refined knowledge load and research productivity in equilibrium.

Along the balanced growth path (BGP), we establish three results under the Pareto optimal allocation. First, as shown in Figure 2, the equilibrium refined knowledge load and research productivity are *nonmonotonic* in the pace of AI progress. Research productivity is maximized at the physiological benchmark at which the human brain attains its peak performance, where knowledge-spillover gains and knowledge-burden costs balance. Starting from a low refined knowledge load, faster AI progress initially raises the knowledge load that reaches human recombination. However, the equilibrium load reaches its highest point before attaining the brain’s physiological benchmark; we call this point the *AI-neutrality point*. Relative to the physiological benchmark, the resulting gap represents a waste of social intelligence: potentially useful knowledge may be

filtered out before it enters human recombination. This waste, however, reflects an intertemporal trade-off between preserving knowledge spillovers and reducing cognitive burden. In the intermediate region, this trade-off generates an *AI trap*: faster AI progress lowers research productivity because spillover losses dominate burden relief.

Second, we propose an elasticity-based criterion to assess whether innovation becomes easier as the AI growth rate increases. We find that faster AI growth makes innovation easier in two regions: a strongly spillover-dominant region, where AI expands the set of knowledge that humans can effectively process enough to sustain spillovers despite selective discarding; and a burden-dominant region, where second-order burden relief outweighs first-order spillover loss. In the weakly spillover-dominant region—the AI trap region—faster AI growth makes innovation harder because filtering erodes spillovers more than it eases burdens, reducing the returns to the brain’s creative thinking.

Finally, faster AI growth raises long-run growth, but its effect on the R&D labor share is ambiguous. When refined knowledge is scarce, faster AI growth expands the set of knowledge that humans can effectively process, strengthens spillovers, and raises research productivity, generating an R&D labor-saving effect. When refined knowledge is abundant, the marginal spillover gains from faster AI progress weaken or turn negative, so the R&D labor share increases.

The rest of the paper is organized as follows. Section 2 reviews the related literature. Section 3 presents the model and characterizes the BGP equilibrium. Section 4 studies the effects of faster AI progress on research productivity, the ease of innovation, and the R&D labor share. Section 5 concludes.

2 Related Literature

Our paper contributes to the literature on the economics of science and innovation. A notable phenomenon in this field is the decline in research productivity over the past decades. For example, Bloom et al. (2020) document substantial declines across sectors such as semiconductors, agriculture, and medicine. Some scholars attribute this decline to the increasing depth and breadth of scientific knowledge. Jones (2009) presents stylized facts showing that the expansion of knowledge raises cognitive burdens on researchers, leading to a shift from individual to team-based innovation, prolonged training periods, and delayed peak productivity. Subsequent studies corroborate the trend of later-life innovation among leading inventors and scientists (Jones, 2010), and document rising co-authorship in economics as a response to increasing

specialization and knowledge complexity (Jones, 2021; Chen et al., 2025), as well as the discouraging effect of the knowledge burden on disruptive and novel innovation (Park et al., 2023; Grashof and Kopka, 2023).¹

From a theoretical perspective, Jones (2009) develops an idea-based growth model in which innovators incur rising costs to assimilate frontier knowledge, leading to prolonged education, greater specialization, and greater reliance on teams. For quantitative analysis, Bloom et al. (2020) propose a simple growth model that decomposes economic growth into two multiplicative factors: research productivity and the number of researchers. This model implies that sustained technological progress—and thus economic growth—requires exponentially increasing R&D investment. In contrast, we develop an endogenous growth model that explicitly incorporates the knowledge burden. Building on Bloom et al. (2020), we further decompose research productivity into two components: a negative effect from the knowledge burden and a positive effect from knowledge spillovers. This tractable framework allows us to examine the effect of AI on innovation and growth through two distinct channels.

Our study advances growth theory from a broader perspective. Since Romer (1990) introduced an endogenous growth model based on the nonrivalry of knowledge, subsequent developments—whether in the expanding-variety framework (e.g., Rivera-Batiz and Romer (1991) and Jones (1995)) or the quality-ladder framework (e.g., Grossman and Helpman (1991a,b) and Aghion and Howitt (1992))—have uniformly assumed purely positive knowledge spillovers. Beyond Jones (2009), who examines the economic implications of knowledge burden within a growth framework, Xie and Yang (2025, 2026) identify frictions in spatial and intertemporal knowledge spillovers—stemming from underdeveloped information carrier technologies—as constraints on innovation and growth. In contrast, to the best of our knowledge, this paper is the first to incorporate both knowledge burden and AI-assisted knowledge filtering into an expanding-variety growth framework, thereby showing how AI can shape long-run growth by changing the knowledge load available for human recombination.

Our work also contributes to the emerging literature on AI and innovation.² Among this literature, empirical studies have linked AI to innovation. Cockburn et al. (2019) document a rapid shift in the U.S.

¹There is also evidence to the contrary. Ando et al. (2025) find that, in U.S. manufacturing, R&D has become more effective at generating productivity-enhancing ideas; they attribute the decline in productivity growth to rising technological rivalry and obsolescence.

²In recent years, rapid advances in and the widespread adoption of AI have spurred growing interest in its economic implications, including employment (Acemoglu and Restrepo, 2018; Acemoglu et al., 2022; Felten et al., 2023; Sun and Zhang, 2025), finance (Babina et al., 2024; Cao et al., 2024), risk and regulation (Jones, 2024; Acemoglu and Lensman, 2024), productivity (Noy and Zhang, 2023; Aghion and Bunel, 2024; Brynjolfsson et al., 2025), economic growth (Aghion et al., 2019; Agrawal et al., 2019; Lu, 2021; Trammell and Korinek, 2023; Bao et al., 2024), and international trade (Goldfarb et al., 2019; Sun and Trefler, 2023).

toward learning-oriented research since 2009, coinciding with breakthroughs in deep learning for tasks such as computer vision.³ Babina et al. (2024) show that AI investment promotes firm growth through product innovation. A few theoretical studies explore the relationship between AI and innovation (Aghion et al., 2019; Agrawal et al., 2019, 2023, 2024; Gans, 2025). Aghion et al. (2019) examine the role of AI as a capital input in idea production, its interaction with research labor, and the resulting implications for economic growth. Agrawal et al. (2019), building on Weitzman (1998), analyze how AI affects search and combination processes within complex knowledge spaces during innovation. Agrawal et al. (2023, 2024) model innovation as a two-stage process—combinatorial prediction and hypothesis testing—and use survival analysis to assess the effects of AI adoption on innovation success probability, search duration, and expected profits. Gans (2025) argues that AI reshapes research incentives by introducing complementarity between scientific novelty and decision-making effectiveness, showing that sufficiently advanced AI tools encourage more novel rather than incremental research. Jones (2025) characterizes AI’s role in R&D as the automation of research tasks. However, macroeconomic theory has paid less attention to AI as a knowledge-filtering technology that changes the burden and spillovers associated with human recombination. We model this channel and show how faster AI progress can raise long-run growth without necessarily making innovation easier.

3 The Model

Our endogenous growth model incorporates both the advantages and limitations of the brain in the innovation process. We focus on equilibrium outcomes under the Pareto optimal allocation.

3.1 Economic Environment

The representative consumer. The economy consists of a constant mass $L > 0$ of homogeneous consumers who supply labor inelastically. Each consumer has constant relative risk aversion (CRRA) preferences. By choosing the consumption path $c(t)$ for $t \in (0, \infty)$, a representative consumer maximizes her discounted

³Cockburn et al. (2019) argue that among the three key AI trajectories—robotics, symbolic systems, and deep learning—only deep learning, due to its general-purpose nature, is likely to transform the innovation process; symbolic systems have stagnated with limited future relevance, and robotics, while capable of substituting for labor, is unlikely to fundamentally reshape innovation.

lifetime utility

$$\int_0^\infty e^{-\rho t} \frac{c(t)^{1-\gamma} - 1}{1-\gamma} dt, \quad (1)$$

where $\rho > 0$ is the subjective discount rate, $\gamma > 0$ is the coefficient of relative risk aversion, and $c(t)$ denotes per capita consumption.

Final goods producers. The final goods market is perfectly competitive. At time t , the production function for the final good is

$$Y(t) = z(t)(1 - s(t))L, \quad (2)$$

where $Y(t)$ denotes the total output of the final good in the economy, $z(t)$ represents the production technology, and $s(t) \in (0, 1)$ indicates the fraction of labor allocated to R&D for production technology. Thus, $1 - s(t)$ is the fraction of labor allocated to final good production. As a result, the final good output per capita, defined as $y(t) \equiv Y(t)/L$, is given by $z(t)(1 - s(t))$.

Production technology. As a downstream technology, the advancement of production technology depends on AI technology, R&D labor input, and the existing knowledge base. At time t , the stock of production knowledge, $N_z(t)$, depends on the level of production technology, $z(t)$, as follows:

$$N_z(t) = d \cdot z(t)^\nu, \quad (3)$$

where $d > 0$ captures the strength, and $\nu \in (0, 1]$ governs the scale effect in converting production technology into knowledge stock.

The brain typically filters knowledge to enable thought. [Licklider \(1960\)](#) argues that approximately 85% of his “thinking” time is spent on clerical or mechanical tasks that prepare for actual thinking, decision-making, or learning, including searching, calculating, plotting, transforming, determining the logical or dynamic consequences of a set of assumptions or hypotheses, and preparing the way for a decision or insight. The renowned communication theorist Marshall McLuhan distinguishes hot from cool media, defining hot media as high definition (the state of being well filled with data), thereby limiting audience participation ([McLuhan, 1994](#)). He argues that high-definition experiences must be forgotten, censored, and cooled before they can be assimilated. Censorship is crucial for learning; without it, unfiltered exposure to shocks would

result in mental collapse. Motivated by these insights, we formalize innovation as a two-stage process. The first stage involves AI technology $A(t)$ processing the stock of production knowledge $N_z(t)$ to generate an effective refined knowledge load $x(t)$ that enters human cognition. This process is represented as

$$x(t) = h(N_z(t), A(t)) \equiv \frac{N_z(t)}{A(t)}, \quad (4)$$

where the function $h(\cdot, \cdot)$ satisfies $\partial h(N_z, A)/\partial N_z > 0$ and $\partial h(N_z, A)/\partial A < 0$. Thus, for a given level of AI technology, a larger stock of production knowledge raises the refined knowledge load; for a given stock of production knowledge, more advanced AI lowers the load that enters human cognition. To make the model tractable, we define this function as $h(N_z, A) \equiv N_z/A$ throughout the paper.⁴

The second stage involves R&D labor expanding the frontier of innovative possibilities in production technology. This process is expressed as:

$$\frac{\dot{z}(t)}{z(t)} = \underbrace{f(x(t))}_{\text{research productivity}} \times \underbrace{s(t)L}_{\text{number of researchers}}, \quad (5)$$

where the function $f(\cdot)$ captures the contribution of each brain to technological innovation. Here, $f(x)$ and sL correspond to research productivity and the number of researchers, respectively, similar to the formulation in Bloom et al. (2020). We specify the function $f(\cdot)$ as follows:

$$f(x(t)) = F_0 \exp \left(\underbrace{-b \left[\ln \frac{x(t)}{x_0} \right]^2}_{\text{knowledge burden}} + \underbrace{a \ln \frac{x(t)}{x_0}}_{\text{knowledge spillovers}} \right), \quad (6)$$

where $F_0 > 0$ is a scale constant, $a > 0$ parameterizes knowledge spillovers, $b > 0$ captures the knowledge burden, and $x_0 > 0$ is a reference level used to normalize $x(t)$. This specification implies the following

⁴One possible microfoundation for the reduced-form mapping $x(t) = N_z(t)/A(t)$ is as follows. Let $\phi(t) \geq 0$ denote the intensity of AI-assisted screening applied to the stock of production knowledge $N_z(t)$, and let $x(t) = N_z(t) \exp(-\phi(t))$ denote the effective knowledge load entering the brain. Assume that, per unit of production knowledge, screening yields a benefit $\phi(t)$ and incurs a cost $(\exp(\phi(t)) - 1)/A(t)$, where a higher level of AI technology $A(t)$ lowers the cost of achieving a given screening intensity. The stage-1 problem is therefore $\max_{\phi(t) \geq 0} N_z(t)\phi(t) - N_z(t)(\exp(\phi(t)) - 1)/A(t)$. Assuming $A(t) \geq 1$, the objective is strictly concave in $\phi(t)$, so the first-order condition $1 - \exp(\phi(t))/A(t) = 0$ is sufficient. It follows that $\phi^*(t) = \ln A(t)$, and hence $x(t) = N_z(t) \exp(-\phi^*(t)) = N_z(t)/A(t)$.

properties:

$$\lim_{x \rightarrow 0^+} f(x) = 0, \quad \lim_{x \rightarrow +\infty} f(x) = 0, \quad \frac{\partial f(x)}{\partial x} > 0 \text{ for } x < \widehat{x}, \quad \frac{\partial f(x)}{\partial x} < 0 \text{ for } x > \widehat{x},$$

where $\widehat{x} \equiv x_0 \exp(\frac{a}{2b})$ is the threshold at which the monotonicity of the function changes and can be interpreted as the level of refined knowledge at which the brain operates at optimal performance. Thus, $f(x)$ is an inverted-U in x , shaped by positive knowledge spillovers and negative knowledge burdens.

AI technology. AI technology is upstream in the innovation process, as shown in Figure 1. AI technology progresses exogenously at a constant rate $m > 0$. Thus, at time t , the level of AI technology is:

$$A(t) = A(0) \exp(mt), \quad (7)$$

where $A(0) > 0$ is the initial level of AI technology. This setup is sufficient to explore the core mechanisms by which AI technology and the brain influence innovation and growth.

Resource constraint. The two resources involved are labor and the final good. The labor constraint has been implicitly accounted for by allocating $1 - s(t)$ of labor to the final good production. Thus, the only remaining resource constraint is the final good constraint:

$$c(t)L = Y(t). \quad (8)$$

3.2 Pareto Optimal Problem

The dynamic optimization problem of a social planner is:

$$\max_{\{c, s, x, z\}} \int_0^\infty e^{-\rho t} \frac{c(t)^{1-\gamma} - 1}{1-\gamma} dt, \quad (9)$$

subject to

$$c(t) = z(t)(1 - s(t)), \quad (10)$$

$$x(t) = \frac{dz(t)^\nu}{A(t)}, \quad (11)$$

$$\dot{z}(t) = z(t)s(t)Lf(x(t)). \quad (12)$$

Here, the combination of equations (2) and (8) leads to equation (10), equation (11) follows from equations (3) and (4), and equation (12) is the equivalent transformation of equation (5). Thus, the current-value Hamiltonian of this maximization problem can be expressed as:

$$\mathcal{H}(s, z, \lambda) = \frac{[z(t)(1 - s(t))]^{1-\gamma} - 1}{1 - \gamma} + \lambda(t)z(t)s(t)Lf(x(t)), \quad (13)$$

where $x(t) = \frac{dz(t)^\gamma}{A(t)}$; $A(t)$ grows exogenously as in equation (7); and $\lambda(t)$ is the current-value costate variable. We explicitly incorporate $f(x(t))$ as defined in equation (6) above.

3.3 Existence of Equilibrium

In the model, the balanced growth path (BGP) equilibrium is defined as one in which the growth rates of per capita consumption, production technology, and per capita output of the final good converge to the same constant value, the labor allocation between final good production and R&D for production technology approaches constant ratios, and the level of refined knowledge tends toward a constant value. Throughout the paper, g_k denotes the growth rate of any variable k , and k^* denotes the equilibrium value of any variable k . The elasticity of research productivity with respect to the refined knowledge load is $\varepsilon(x^*) := f'(x^*)x^*/f(x^*)$, which captures the local return to increasing the knowledge that enters human cognition.

For model tractability, we impose two technical assumptions on the parameters. The first rules out a corner solution for the fraction of labor allocated to R&D versus production, while the latter ensures the transversality condition and streamlines the equilibrium characterization. These assumptions are maintained throughout the subsequent analysis.

Assumption 1. $f(x_{\min}) > \frac{m}{\nu L}$ where $x_{\min} \equiv x_0 \exp[\frac{a}{2b} - \frac{\rho + (\gamma-1)m/\nu}{2bm}]$.

Assumption 2. $\rho\nu > m$ and $\min\{a, 2\sqrt{2}\sqrt{b}\} > \frac{\rho}{m} + \frac{\gamma}{\nu}$.

For later reference, let $x_{\max}$ denote the larger solution to $f(x) = m/(\nu L)$, so that $x_{\max}$ is the upper refined-knowledge level compatible with an interior R&D labor share.

Next, we analyze the BGP equilibrium, as outlined in the following theorem and propositions. All proofs are provided in the Appendix.

Theorem 1. *There exists a unique interior BGP equilibrium such that $g_c^* = g_z^* = g_y^* = \frac{m}{\nu}$ and $s^* = \frac{m}{L\nu f(x^*)}$, where x^* solves $\frac{m}{\nu} = \frac{1}{\gamma}[Lf(x^*) + m\varepsilon(x^*) - \rho]$.*

Theorem 1 demonstrates that the model admits a unique BGP equilibrium. Specifically, the BGP growth rates of per capita consumption, production technology, and per capita output are identical, with common value m/ν , determined by the AI technology progress rate m and the parameter ν (which captures the scale effect of converting production technology into knowledge). Faster AI progress supports faster growth in productive knowledge, consistent with the BGP equilibrium in which the refined-knowledge level remains constant. The common BGP growth rate is pinned down by the stationarity of the refined knowledge load. Since $x(t) = dz(t)^\nu / A(t)$ and $A(t)$ grows at rate m , a constant x^* along the BGP requires $g_z^* = m/\nu$. Thus, faster AI progress raises long-run growth through the stationarity condition. The burden–spillover trade-off instead determines the equilibrium refined knowledge load, research productivity, and the R&D labor share through $m/\nu = [Lf(x^*) + m\varepsilon(x^*) - \rho] / \gamma$, together with $s^* = m/[L\nu f(x^*)]$.

4 Effects of Faster AI Progress

4.1 On Refined Knowledge Load and Research Productivity

Proposition 1 characterizes how the equilibrium refined knowledge load, x^* , responds to changes in the AI growth rate, m , as illustrated in Figure 3. When x^* is sufficiently low, accelerated progress in AI technology increases the equilibrium refined knowledge load in the second stage of the innovation process—the creative thinking stage—and thus enhances research productivity (since x^* lies in the region where knowledge spillovers dominate, i.e., $x^* < \widehat{x}$). In contrast, when x^* is sufficiently high, faster AI progress reduces the equilibrium refined knowledge load.

Proposition 1. *When $x^* < \widetilde{x} \equiv x_0 \exp(\frac{a-\gamma/\nu}{2b})$, the equilibrium refined knowledge load, x^* , is increasing with the AI growth rate m . When $x^* > \widetilde{x}$, x^* is decreasing in m . At $x^* = \widetilde{x}$, AI neutrality obtains: any change in m does not alter x^* .*

In particular, when $\widetilde{x} < x^* < \widehat{x}$, additional refined knowledge would still raise research productivity in a partial-equilibrium sense. However, faster AI progress reduces the equilibrium knowledge load that reaches human recombination. We refer to this region as the “AI trap” region. In this region, spillover losses from a lower knowledge load dominate burden relief, so research productivity falls even though x^* remains below $\widehat{x}$, the physiological benchmark at which the human brain achieves peak research productivity. At $x^* = \widetilde{x}$, AI neutrality obtains: a marginal change in the AI growth rate has no first-order effect on equilibrium refined

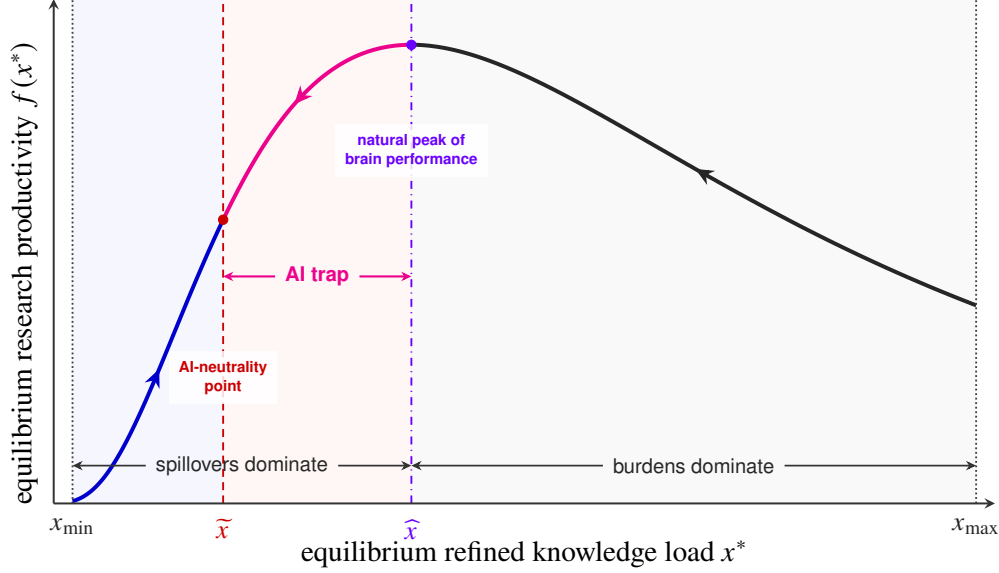


Figure 3: Effects of Faster AI Progress on Refined Knowledge Load and Research Productivity

knowledge and hence no first-order effect on research productivity. Overall, the nonmonotonic response of refined knowledge—and thus research productivity—to faster AI growth reveals a waste of social intelligence relative to the brain’s physiological productivity benchmark. This waste is not a market distortion; it reflects the planner’s trade-off between the spillover value of additional knowledge and the cognitive burden of processing it.

4.2 On the Ease of Innovation

As AI technology advances, is innovation becoming easier? Higher growth is not the same as easier innovation. To distinguish the two, we propose a criterion based on relative elasticities:

$$\varepsilon_{g^*,m} \equiv \frac{\partial g^*/g^*}{\partial m/m} > \varepsilon_{s^*,m} \equiv \frac{\partial s^*/s^*}{\partial m/m}, \quad (14)$$

where $\varepsilon_{g^*,m}$ represents the elasticity of g^* with respect to m , and $\varepsilon_{s^*,m}$ represents the elasticity of s^* with respect to m . The relation $g^* = f(x^*) \times s^*L$ implies that $\varepsilon_{g^*,m} - \varepsilon_{s^*,m}$ corresponds to the elasticity of research productivity with respect to m . The criterion that AI technology makes innovation easier can also be equivalently interpreted as the elasticity of research productivity $f(x^*)$ with respect to m being positive. Thus, we obtain the following proposition.

Proposition 2. *AI technology makes innovation easier (i.e., $\varepsilon_{g^*,m} > \varepsilon_{s^*,m}$) if either $x^* < \tilde{x}$ or $x^* > \hat{x}$; AI*

technology makes innovation harder (i.e., $\varepsilon_{g^*,m} < \varepsilon_{s^*,m}$) if $\tilde{x} < x^* < \hat{x}$.

The proposition characterizes the conditions under which AI technology makes innovation easier. Whether AI facilitates innovation depends on both (i) the relative importance of the knowledge burden versus knowledge spillovers for research productivity and (ii) how the equilibrium refined knowledge load responds to faster AI growth. We identify two cases in which AI makes innovation easier. First, when knowledge spillovers substantially dominate the knowledge burden (i.e., $x^* < \tilde{x}$), faster AI growth enables humans to process more knowledge, thereby allowing these spillovers to be realized despite selective knowledge discarding. Second, when the knowledge burden dominates spillovers (i.e., $x^* > \hat{x}$), faster AI growth reduces the knowledge burden through a second-order channel; this higher-order relief dominates the first-order spillover loss.

In contrast, when $\tilde{x} < x^* < \hat{x}$, the knowledge-burden relief from faster AI growth is insufficient to offset the substantial weakening of knowledge spillovers; thus, faster AI progress makes innovation harder. The intuition is that AI diminishes knowledge spillovers significantly—due to the selective discarding of knowledge—relative to its effect on easing the knowledge burden during the innovation process. This mechanism is especially relevant for cross-domain innovation, where ideas that appear irrelevant in one field may become useful inputs for recombination in another. Human researchers may preserve and exchange such seemingly “wasteful” knowledge through informal communication, broad reading, and cross-field collaboration. Here, “wasteful” does not mean socially valueless; rather, it refers to knowledge that appears to have low immediate task relevance but retains option value for future recombination. By contrast, specialized AI systems may screen out information classified as low relevance for the immediate task. As AI filtering becomes more intensive, serendipitous recombination may become harder if peripheral knowledge and cross-field insights are less likely to reach researchers.

4.3 On R&D Labor Share

Proposition 3. *The proportion of labor allocated to the R&D sector decreases with the growth rate of AI technology if $x^* < x_0 \exp(\frac{a}{2b} - \frac{m}{\rho})$ and increases if $x^* > x_0 \exp(\frac{a}{2b} - \frac{m}{\rho})$.*

The proposition has two implications. First, when the equilibrium level of refined knowledge required for the brain’s creative thinking is sufficiently low (i.e., $x^* < x_0 \exp(\frac{a}{2b} - \frac{m}{\rho})$, for convenience, let $\bar{x} \equiv x_0 \exp(\frac{a}{2b} - \frac{m}{\rho})$), faster AI growth expands the amount of knowledge that humans can process during innovation. This

increases the equilibrium refined-knowledge level and strengthens knowledge spillovers, thereby significantly raising research productivity and generating R&D labor-saving effects (note that Assumption 2 implies $\bar{x} < \tilde{x}$). Second, when the marginal knowledge spillovers induced by faster AI growth are weak for $\bar{x} < x^* < \tilde{x}$, decline for $\tilde{x} < x^* < \hat{x}$, or when faster AI growth primarily alleviates the knowledge burden for $x^* > \hat{x}$, the social planner allocates a higher fraction of labor to R&D to offset lower research productivity or insufficient marginal gains in research productivity.

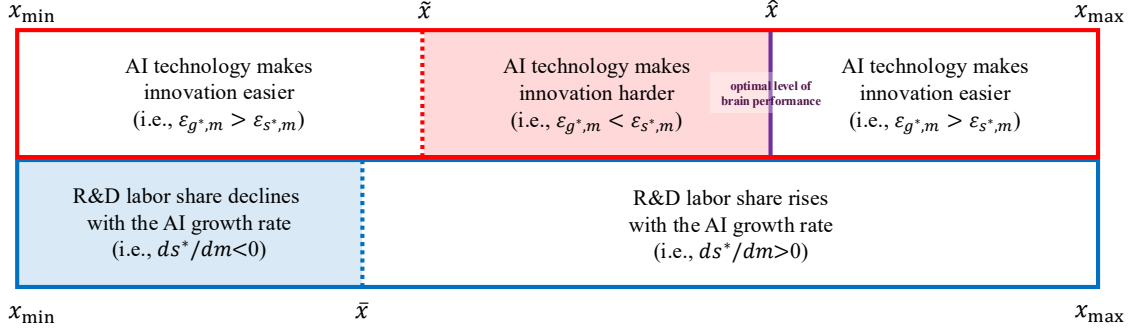


Figure 4: Effects of AI Technological Progress on R&D Performance over Different Ranges of x^*

The equilibrium implications stated in Propositions 2 and 3 are summarized in Figure 4.⁵ In summary, whether AI progress makes innovation harder or easier need not induce a monotonic change in the R&D labor share. When equilibrium refined knowledge load is very low, faster AI growth raises it and facilitates innovation. Yet because research productivity increases by more than the AI growth rate, the marginal returns to R&D labor can fall below those to labor in the final good production, reducing the R&D labor share. Outside this region, once equilibrium refined knowledge load is sufficiently high, allocating a higher fraction of labor to R&D becomes more profitable, regardless of whether AI makes innovation harder or easier.

5 Conclusion

In this paper, we develop an endogenous growth framework in which AI filters accumulated knowledge before bounded human cognition recombines it into new ideas. This structure decomposes research productivity, as characterized in Bloom et al. (2020), into positive knowledge spillovers and negative knowledge

⁵The equilibrium in Figure 4 is characterized by $b < \frac{1}{2} \left(\frac{\rho}{m} + \frac{\gamma-1}{\nu} \right) \frac{\rho}{m}$. If instead $b > \frac{1}{2} \left(\frac{\rho}{m} + \frac{\gamma-1}{\nu} \right) \frac{\rho}{m}$ —which implies $\bar{x} < x_{\min}$ —then $ds^*/dm > 0$ always holds. We therefore focus on the former case.

burdens, and it delivers three main results. First, faster AI growth raises the long-run growth rate, but does not necessarily make innovation easier. Second, the equilibrium refined knowledge load need not coincide with the load that maximizes the brain’s physiological research productivity, implying a waste of social intelligence relative to this physiological benchmark rather than a market distortion. Third, AI makes innovation easier when spillovers survive filtering or burden relief dominates, but harder in the intermediate region where spillover loss outweighs burden reduction; its effect on the fraction of labor allocated to R&D is therefore ambiguous.

The model also generates testable empirical predictions. AI adoption should raise research productivity when usable knowledge is scarce or when information overload is severe, but stronger filtering may reduce novelty, atypical combinations, and cross-domain recombination in intermediate knowledge environments. These predictions can be examined using patent texts, scientific publications, grant proposals, and researcher-level measures of AI-assisted R&D. They also suggest that AI systems used for discovery should not be evaluated only by speed or task-specific relevance: systems that preserve some diversity and peripheral exposure may reduce knowledge burdens without eliminating the spillovers that support creative recombination.

The broader implication is that AI creates value not merely by processing more knowledge, but by organizing knowledge in ways that preserve the conditions for human creativity. Standard growth models emphasize the nonrivalry of knowledge; our framework separates the existence of knowledge from its cognitive usability. A society can therefore accumulate more knowledge and deploy better AI while still leaving some recombination potential unused relative to the brain’s physiological productivity benchmark if filtering leaves researchers with too narrow a basis for discovery. Future work can extend the framework to decentralized incentives, endogenous AI investment, heterogeneous researchers, and AI systems that affect not only search but also the second-stage creative process itself.

Appendix: Proofs

We introduce some notation that will simplify the analysis in the following proofs. Let $u(t) := \ln \frac{x(t)}{x_0}$, $\psi(u(t)) := au(t) - b[u(t)]^2$, and $\mathcal{F}(u(t)) := F_0 \exp[\psi(u(t))]$. That is, we can rewrite $f(x(t))$ as $\mathcal{F}(u(t))$. Note that the mapping between x^* and u^* is bijective and strictly increasing. Assumption 1 can also be restated as

Assumption 1'. $\mathcal{F}(u_{\min}) > \frac{m}{\nu L}$ where $u_{\min} = \frac{a}{2b} - \frac{\rho + (\gamma - 1)m/\nu}{2bm}$.

We can rewrite the elasticity of research productivity concerning refined knowledge as $\mathcal{E}(u(t)) := a - 2bu(t)$.

A Proof of Theorem 1

Proof. The social planner's maximization problem, characterized by the current-value Hamiltonian in equation (13), implies that the time paths of $s(t)$, $z(t)$, and $\lambda(t)$ must satisfy the following first-order necessary conditions:

$$\mathcal{H}_s(s, z, \lambda) = -z(t)[z(t)(1 - s(t))]^{-\gamma} + \lambda(t)z(t)Lf(x(t)) = 0, \quad (\text{O1})$$

$$\mathcal{H}_z(s, z, \lambda) = (1 - s(t))[z(t)(1 - s(t))]^{-\gamma} + \lambda(t)s(t)L[f(x(t)) + \nu f'(x(t))x(t)] = \rho\lambda(t) - \dot{\lambda}(t), \quad (\text{O2})$$

$$\dot{z}(t) = \mathcal{H}_\lambda(s, z, \lambda) = z(t)s(t)Lf(x(t)). \quad (\text{O3})$$

The transversality condition is $\lim_{t \rightarrow \infty} [\exp(-\rho t)\lambda(t)z(t)] = 0$.

To prepare for the equilibrium analysis, we rearrange the above conditions as follows. First, when expressed in growth rates, equation (O1) yields

$$-\gamma \left[g_z(t) + \frac{-\dot{s}(t)}{1 - s(t)} \right] = g_\lambda(t) + \frac{\dot{f}(x(t))}{f(x(t))}. \quad (\text{O4})$$

Equation (O1) also implies

$$\frac{[z(t)(1 - s(t))]^{-\gamma}}{\lambda(t)} = Lf(x(t)). \quad (\text{O5})$$

Second, incorporating equation (O5) into equation (O2) gives

$$Lf(x(t)) + \nu s(t)Lf'(x(t))x(t) = \rho - g_\lambda(t). \quad (\text{O6})$$

Third, equation (O3) directly implies

$$g_z(t) = s(t)Lf(x(t)). \quad (\text{O7})$$

Next, we determine the BGP equilibrium. Recall the definition of the BGP equilibrium. As $t \rightarrow \infty$, $x(t)$ converges to a constant value x^* . Therefore, equation (11) implies $g_z^* = \frac{m}{\nu}$. In the BGP equilibrium, the fraction of labor allocated to R&D tends to a constant value s^* , which, by condition (10), implies $g_c^* = g_z^*$. Since $y(t) = z(t)(1 - s(t))$, it follows that $g_y^* = g_z^*$. Taking into account equation (O4) in BGP equilibrium, we obtain: $g_c^* = g_z^* = g_y^* = \frac{m}{\nu}$ and $g_\lambda^* = -\frac{\gamma m}{\nu}$. It follows immediately that $g^* \equiv g_c^* = g_z^* = g_y^*$ increases in m . Thus, condition (O7) yields the fraction of labor allocated to the R&D sector in the BGP equilibrium as $s^* = \frac{m}{L\nu f(x^*)}$. Moreover, substituting the full expressions for s^* and g_λ^* into equation (O6) yields

$$\frac{m}{\nu} = \frac{1}{\gamma} [Lf(x^*) + m\varepsilon(x^*) - \rho]. \quad (\text{O8})$$

Thus, once x^* is obtained from this equation, all equilibrium outcomes are determined. It remains to prove that equation (O8) admits at least one solution for x^* .

We further define $x_l := \min\{x \mid f(x) = \frac{m}{L\nu}\}$ and $x_{\max} := \max\{x \mid f(x) = \frac{m}{L\nu}\}$. To ensure that s^* lies within $(0, 1)$, it must hold that $f(x^*) > \frac{m}{L\nu}$, or equivalently, $L\mathcal{F}(u^*) > m/\nu$. This condition implies $x_l < x^* < x_{\max}$, or equivalently, $u_l < u^* < u_{\max}$, where $u_l = u(x_l)$ and $u_{\max} = u(x_{\max})$. The existence of x^* is equivalent to that of u^* , as the mapping between them is bijective and strictly increasing. Equation (O8) can be rewritten as

$$L\mathcal{F}(u^*) = -m\varepsilon(u^*) + \gamma m/\nu + \rho. \quad (\text{O9})$$

Define $\phi(u; m) \equiv L\mathcal{F}(u) + m\varepsilon(u) - \gamma m/\nu - \rho$. Then equation (O9) is equivalent to $\phi(u; m) = 0$. Let $h(u) \equiv -m\varepsilon(u) + \gamma m/\nu + \rho$. By construction, $h(u_{\min}) = m/\nu$. Assumption 1' implies $L\mathcal{F}(u_{\min}) > m/\nu$, and hence $\phi(u_{\min}; m) > 0$. Moreover, by the definition of $u_{\max}$, $L\mathcal{F}(u_{\max}) = m/\nu$. Since $h(u)$ is strictly increasing and $u_{\max} > u_{\min}$, we have $h(u_{\max}) > m/\nu$, and therefore $\phi(u_{\max}; m) < 0$. Hence, by continuity,

there exists at least one solution $u^* \in (u_{\min}, u_{\max})$. It remains to show uniqueness. At any solution to $\phi(u; m) = 0$, equation (O9) implies $L\mathcal{F}(u) = -m\mathcal{E}(u) + \frac{\gamma m}{\nu} + \rho$. Using $\mathcal{E}'(u) = -2b$, we obtain $\phi_u(u; m) = L\mathcal{F}(u)\mathcal{E}(u) - 2bm = -m \left[\mathcal{E}(u)^2 - \left(\frac{\gamma}{\nu} + \frac{\rho}{m} \right) \mathcal{E}(u) + 2b \right]$. Assumption 2 implies $\left(\frac{\gamma}{\nu} + \frac{\rho}{m} \right)^2 < 8b$, so the quadratic term in brackets is strictly positive for all u . Thus, $\phi_u(u; m) < 0$ at every solution. Since $\phi(u_{\min}; m) > 0$ and $\phi(u_{\max}; m) < 0$, and every zero of ϕ is crossed from above to below, there can be only one zero in $(u_{\min}, u_{\max})$. Therefore, equation (O9) admits a unique solution u^* , and the strictly increasing mapping between u^* and x^* implies a unique x^* satisfying equation (O8). $\square$

B Proof of Proposition 1

Proof. As established in the proof of Theorem 1, equation (O9) holds in equilibrium. We have that

$$u_{\min} = \frac{a}{2b} - \frac{\frac{\rho}{m} + \frac{\gamma-1}{\nu}}{2b} < u^* < u_{\max} = \frac{a}{2b} + \frac{\sqrt{a^2 - 4b \ln \left(\frac{m}{\nu} \frac{1}{LF_0} \right)}}{2b}.$$

Let $\phi := L\mathcal{F}(u^*) + m\mathcal{E}(u^*) - \gamma m/\nu - \rho = 0$. By the implicit function theorem, we have $\frac{du^*}{dm} = -\frac{\partial \phi}{\partial m} \Big/ \frac{\partial \phi}{\partial u^*}$ where the partial derivatives are

$$\frac{\partial \phi}{\partial m} = (a - 2bu^*) - \gamma/\nu \text{ and } \frac{\partial \phi}{\partial u^*} = LF_0 \exp(au^* - bu^{*2})(a - 2bu^*) - 2bm.$$

Thus, we obtain

$$\frac{du^*}{dm} = \frac{2b \left(u^* - \frac{a-\gamma/\nu}{2b} \right)}{LF_0 \exp(au^* - bu^{*2})(a - 2bu^*) - 2bm}.$$

Using the equilibrium condition from (O9), the derivative becomes

$$\frac{du^*}{dm} = \frac{2b \left(u^* - \frac{a-\gamma/\nu}{2b} \right)}{-4b^2 mu^{*2} + 2b(2am - \gamma m/\nu - \rho)u^* + [m(a\gamma/\nu - a^2 - 2b) + a\rho]}. \quad (\text{O10})$$

Next, we examine the sign of $\frac{du^*}{dm}$. First, it is evident that if $u^* > \frac{a-\gamma/\nu}{2b}$, the numerator is positive; if $u^* < \frac{a-\gamma/\nu}{2b}$, it is negative; and if $u^* = \frac{a-\gamma/\nu}{2b}$, AI neutrality obtains, so u^* (and hence x^*) does not vary with m . Second, the denominator remains strictly negative because $2\sqrt{2}\sqrt{b} > \frac{\gamma}{\nu} + \frac{\rho}{m}$ in Assumption 2. Thus, we

can conclude that: if $\frac{a-\gamma/\nu}{2b} < u^* < u_{\max}$, there is $\frac{du^*}{dm} < 0$; if $u_{\min} < u^* < \frac{a-\gamma/\nu}{2b}$, there is $\frac{du^*}{dm} > 0$. Moreover, since $u^* = \ln(\frac{x^*}{x_0})$, it follows that

$$\frac{du^*}{dm} = \underbrace{\frac{1}{x^*}}_{>0} \frac{dx^*}{dm}, \quad (\text{O11})$$

which implies $\text{sign}\left\{\frac{dx^*}{dm}\right\} = \text{sign}\left\{\frac{du^*}{dm}\right\}$, establishing the result. $\square$

C Proof of Proposition 2

Proof. Recall the equilibrium results: $g^* = \frac{m}{\nu}$ and $s^* = \frac{m}{L\nu f(x^*)}$. The definitions of elasticities imply:

$$\varepsilon_{g^*,m} = 1 \text{ and } \varepsilon_{s^*,m} = 1 - m \frac{f'(x^*)}{f(x^*)} \frac{dx^*}{dm} = 1 - m\mathcal{E}(u^*) \frac{du^*}{dm}.$$

where the last equality follows from (O11). We have

$$\begin{aligned} \Delta E &:= \varepsilon_{g^*,m} - \varepsilon_{s^*,m} \\ &= m\mathcal{E}(u^*) \frac{du^*}{dm} \\ &= \frac{m(a - 2bu^*)(\frac{\gamma}{\nu} - (a - 2bu^*))}{LF_0 \exp[au^* - bu^{*2}](a - 2bu^*) - 2bm} \end{aligned}$$

where the last equality uses (O10). Therefore,

$$\text{sign}\{\Delta E\} = \text{sign}\{\mathcal{E}(u^*)\} \times \text{sign}\left\{\frac{du^*}{dm}\right\}$$

which gives us $\Delta E > 0$ if $u^* < \frac{a-\gamma/\nu}{2b}$ or $u^* > \hat{u} \equiv a/(2b)$. Otherwise, $\Delta E < 0$. $\square$

D Proof of Proposition 3

Proof. Based on the expression of s^* in BGP equilibrium, we obtain

$$\frac{ds^*}{dm} = \underbrace{\frac{1}{Lv f(x^*)}}_{>0} \left(\underbrace{1 - m \frac{f'(x^*)}{f(x^*)} \frac{dx^*}{dm}}_{=\mathcal{E}_{s^*,m}} \right) < 0 \iff \mathcal{E}_{s^*,m} < 0.$$

We have

$$\begin{aligned} \mathcal{E}_{s^*,m} &= 1 - m\mathcal{E}(u^*) \frac{du^*}{dm} \\ &= 1 - \frac{m\mathcal{E}(u^*)\left(\frac{\gamma}{v} - (a - 2bu^*)\right)}{LF_0 \exp[au^* - bu^{*2}](a - 2bu^*) - 2bm} \\ &= 1 - \frac{m\mathcal{E}(u^*)\left(\frac{\gamma}{v} - \mathcal{E}(u^*)\right)}{L\mathcal{F}(u^*)\mathcal{E}(u^*) + m\mathcal{E}'(u^*)} \\ &= \frac{\mathcal{E}(u^*)(L\mathcal{F}(u^*) + m\mathcal{E}(u^*) - \frac{\gamma m}{v}) + m\mathcal{E}'(u^*)}{L\mathcal{F}(u^*)\mathcal{E}(u^*) + m\mathcal{E}'(u^*)} \\ &= \frac{\rho\mathcal{E}(u^*) + m\mathcal{E}'(u^*)}{L\mathcal{F}(u^*)\mathcal{E}(u^*) + m\mathcal{E}'(u^*)} \end{aligned}$$

where the first line follows from the proof of Proposition 2 and the last equality comes from (O9). Therefore,

$$\frac{ds^*}{dm} < 0 \iff \frac{\rho\mathcal{E}(u^*) + m\mathcal{E}'(u^*)}{L\mathcal{F}(u^*)\mathcal{E}(u^*) + m\mathcal{E}'(u^*)} < 0.$$

Since the denominator is negative (see the proof of Proposition 1), $\frac{ds^*}{dm} < 0$ if and only if the numerator is positive, which is equivalent to $u^* < \frac{a}{2b} - \frac{m}{\rho}$. $\square$

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